

Continuing Professional Development Log

Name: Chi Tat Ip

Current and recent CPD activity:

CPD Activity Title	Description	Dates	CPD Hours
FPGA and HDL Fundamentals	Studying FPGA architectures and hands-on tutorials using SystemVerilog to implement processors. Extended through the project to cover the Xilinx 7-series	Nov 2024 – Mar 2026	70
RISC-V Architecture Overview	Self-study of RISC-V ISA and instruction formats. Used as the conceptual basis for the custom 20-bit SIMD ISA designed for this project.	Nov 2024 – Jan 2026	55
Git and Version Control	Use of Git for source control of HDL, testbenches and the report. Adopted a remote-first workflow after the laptop failure, with daily pushes and tagged milestone commits.	Jun 2025 – Apr 2026	20
HDL Simulation & Verification (Vivado)	Writing self-checking SystemVerilog testbenchs and waveform debugging in Vivado xsim.	Nov 2024 – Apr 2026	75
Bit-Serial Datapath Design	Studied and implemented bit-serial ALU, register file and shift logic. Investigated trade-offs between throughput, area and routing density compared with bit-parallel designs.	Oct 2025 – Jan 2026	40
SIMD / Array Processor Architecture	Studied NEWS interconnects, broadcast control distribution, and more used in this project's 2D array.	Oct 2025 – Feb 2026	35
FPGA Synthesis, P&R and Timing Closure	Hands-on experience with Vivado synthesis, place-and-route, timing reports and floorplanning.	Jan 2026 – Apr 2026	30
CNN Fundamentals & Quantisation for Inference	Self-study of CNN layer types, MNIST topology selection, post-training 8-bit quantisation and the interaction between hardware rounding/saturation and inference accuracy.	Feb 2026 – Apr 2026	35
Technical Writing & LaTeX	Used LaTeX for the final report and Visio for professional architecture diagrams.	Feb 2026 – Apr 2026	20
Project Management Tools	Maintained the Risk Register, Gantt chart and CPD log throughout the project.	Oct 2025 – Apr 2026	15
Total recorded hours			425

Planned and Future CPD activity:

CPD Activity Title	Description	Skills addressed	Dates
Power, Timing and Thermal Analysis (extended)	Extend FPGA work to include detailed power profiling and thermal characterisation.	Power optimisation; thermal management;	May 2026 – Aug 2026
ASIC Design Flow (OPENLane)	Self-study of an industrial ASIC flow to understand how the bit-serial array would migrate from FPGA to silicon.	Industrial EDA flow; standard-cell design;	May 2026 – Sep 2026
High-Level Synthesis (HLS)	Learn Vitis HLS for more advanced development on the FPGA	HLS toolchains; C-to-RTL transformation;	Jun 2026 – Sep 2026
Advanced CNN Architectures and Quantisation	Study quantisation-aware training, mixed-precision inference, and modern lightweight architectures (MobileNet, etc.)	Modern ML inference; hardware-software co-design	Jun 2026 – Dec 2026
UVM-Based Functional Verification	Learn the Universal Verification Methodology to apply constrained-random verification	SystemVerilog OOP; UVM agents/sequences	Jul 2026 – Dec 2026